

Q.1 Start a repository

```
rushi@rushi:~$ mkdir myproject && cd myproject/
rushi@rushi:~/myproject$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/rushi/myproject/.git/
rushi@rushi:~/myproject$ nano index.html
rushi@rushi:~/myproject$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    index.html

nothing added to commit but untracked files present (use "git add" to track)
rushi@rushi:~/myproject$ git add index.html
rushi@rushi:~/myproject$ git commit -m "first commit"
[master (root-commit) 48a75da] first commit
 1 file changed, 88 insertions(+)
 create mode 100644 index.html
rushi@rushi:~/myproject$ git log --one line
fatal: ambiguous argument 'line': unknown revision or path not in the working tree
.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'
rushi@rushi:~/myproject$ git log --oneline
48a75da (HEAD -> master) first commit
rushi@rushi:~/myproject$
```

Q2. The change that was not saved

```
rushi@rushi: ~/myproject × rushi@rushi: ~/myproject
rush i@rush i:~/myproject$ nano index.html
rush i@rush i:~/myproject$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html

no changes added to commit (use "git add" and/or "git commit -a")
rush i@rush i:~/myproject$ git add index.html
rush i@rush i:~/myproject$ git commit -m "updated the page"
[master 65fbf69] updated the page
 1 file changed, 1 insertion(+), 1 deletion(-)
rush i@rush i:~/myproject$ git log --oneline
65fbf69 (HEAD -> master) updated the page
48a75da first commit
rush i@rush i:~/myproject$
```

Q.3 Read the three states

```
rushi@rushi: ~/myproject  ×  rushi@rushi: ~/myproject  ×  rushi@rushi: ~/myproject
about.html  index.html  style.css
rushi@rushi:~/myproject$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    about.html
    style.css

nothing added to commit but untracked files present (use "git add" to track)
rushi@rushi:~/myproject$ git add about.html
rushi@rushi:~/myproject$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   about.html

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    style.css

rushi@rushi:~/myproject$ nano index.html
rushi@rushi:~/myproject$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   about.html

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   index.html

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    style.css

rushi@rushi:~/myproject$
```

Q4. Keeping Secretes Out

```
rushi@rushi:~/myproject$ nano secrete.env
rushi@rushi:~/myproject$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   about.html

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   index.html

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    secrete.env
    style.css

rushi@rushi:~/myproject$ nano .gitignore
rushi@rushi:~/myproject$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
    new file:   about.html

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   index.html

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    .gitignore
    style.css
```

```
rushi@rushi:~/myproject$ git add .gitignore
rushi@rushi:~/myproject$ git commit -m "add gitignore"
[master 535035a] add gitignore
 2 files changed, 2 insertions(+)
 create mode 100644 .gitignore
 create mode 100644 about.html
rushi@rushi:~/myproject$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   index.html

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        style.css

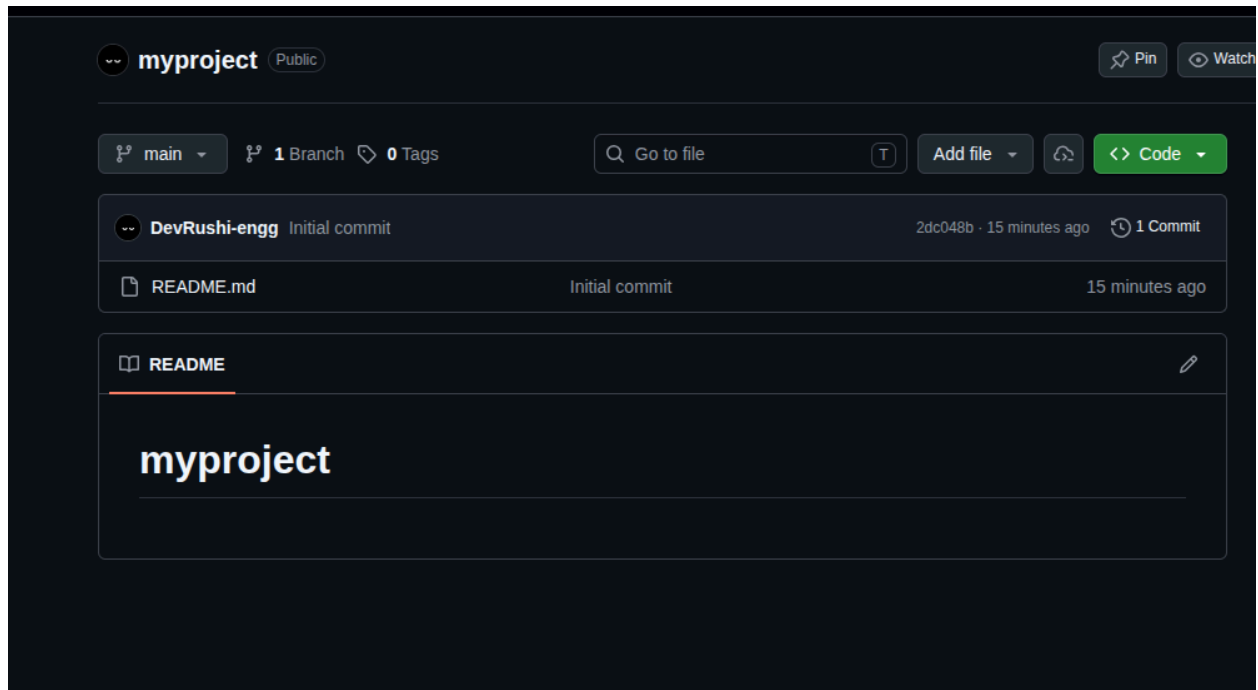
no changes added to commit (use "git add" and/or "git commit -a")
```

Q.5 The push with nowhere to go

```
rushi@rushi: ~/... ×  rushi@rushi: ~/... ×  rushi@rushi: ~/... ×  rushi@rushi: ~/... ×  rushi@rushi: ~/... ×  ✓  
rushi@rushi:~/myproject$ git remote add origin git@github.com:DevRushi-engg/myproject.git
```

```
rushi@rushi:~/myproject$ git push -u origin main  
Enumerating objects: 16, done.  
Counting objects: 100% (16/16), done.  
Delta compression using up to 16 threads  
Compressing objects: 100% (13/13), done.  
Writing objects: 100% (15/15), 2.20 KiB | 1.10 MiB/s, done.  
Total 15 (delta 4), reused 0 (delta 0), pack-reused 0  
remote: Resolving deltas: 100% (4/4), done.  
To github.com:DevRushi-engg/myproject.git  
    2dc048b..fc474af  main -> main  
branch 'main' set up to track 'origin/main'.  
rushi@rushi:~/myproject$
```

Q6. Connect to Github



```
rushi@rushi: ~/... x  rushi@rushi: ~/... x  rushi@rushi: ~/... x  rushi@rushi: ~/... x  rushi@rushi: ~/... x  v
rushi@rushi:~/myproject$ git remote add origin git@github.com:DevRushi-engg/myproject.git
```

```
rushi@rushi:~/myproject$ git push -u origin main
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 16 threads
Compressing objects: 100% (13/13), done.
Writing objects: 100% (15/15), 2.20 KiB | 1.10 MiB/s, done.
Total 15 (delta 4), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (4/4), done.
To github.com:DevRushi-engg/myproject.git
   2dc048b..fc474af  main -> main
branch 'main' set up to track 'origin/main'.
rushi@rushi:~/myproject$
```

main

1 Branch 0 Tags

Go to file



Add file



Code

DevRushi-engg add style.css

fc474af · 2 minutes ago 6 Commits

.gitignore	add gitignore	2 minutes ago
README.md	Initial commit	16 minutes ago
about.html	add gitignore	2 minutes ago
index.html	updated the page	2 minutes ago
style.css	add style.css	2 minutes ago

README



myproject

Q7. The branch that was not there

```
rushi@rushi:~/myproject$ git push -u origin master
error: src refspec master does not match any
error: failed to push some refs to 'github.com:DevRushi-engg/myproject.git'
rushi@rushi:~/myproject$ git branch --show-current
main
rushi@rushi:~/myproject$ git push -u origin master
error: src refspec master does not match any
error: failed to push some refs to 'github.com:DevRushi-engg/myproject.git'
rushi@rushi:~/myproject$ git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
rushi@rushi:~/myproject$
```

Q8 . The full Cycle

```
rushi@rushi:~/myproject$ nano index.html
rushi@rushi:~/myproject$ nano contact.html
rushi@rushi:~/myproject$ git add .
rushi@rushi:~/myproject$ git commit -m "add contact page"
[main d824e2d] add contact page
 2 files changed, 13 insertions(+), 1 deletion(-)
 create mode 100644 contact.html
rushi@rushi:~/myproject$ git push
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 16 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 657 bytes | 657.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:DevRushi-engg/myproject.git
   fc474af..d824e2d  main -> main
rushi@rushi:~/myproject$ git log --oneline
d824e2d (HEAD -> main, origin/main) add contact page
fc474af add style.css
8cbd00e updated the page
50f23c0 add gitignore
7458f7c updated the page
b80c44b first commit
2dc048b Initial commit
rushi@rushi:~/myproject$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
rushi@rushi:~/myproject$
```